benchmark-hwm-chees

A reproducible benchmark of PEACHES (ChEES-adapted Hamiltonian walk move), `emcee`, and NUTS

Abstract

PEACHES is the *Hamiltonian walk move* — an affine-invariant ensemble HMC proposal whose metric is the empirical covariance of the live ensemble — with its integration length tuned by the **ChEES** criterion (Change in the Estimator of the Expected Square; Hoffman, Radul & Sountsov, AISTATS 2021) and its step size by Nesterov dual averaging (Hoffman & Gelman, JMLR 2014). We benchmark it against `emcee` (the gradient-free affine-invariant stretch move; Goodman & Weare 2010, Foreman-Mackey et al. 2013) and NUTS (`numpyro`) on controlled targets with exact moments, over three axes: dimensional scaling, conditioning vs. curvature, and warmup robustness. Every run is checked for convergence (rank-normalized split $\hat{R}$); all configurations and commands are given for exact reproduction.

1 Methods

peaches: a ChEES-adapted Hamiltonian walk move. Let the ensemble be W walkers $q_{1:W} \in \mathbb{R}^d$ with target $\pi(q) \propto e^{-U(q)}$. The *Hamiltonian walk move* ("h-walk") proposes a Hamiltonian trajectory for a subset of walkers using a momentum that lives in the span of the *complement* ensemble: with C the centered complement walkers (rows $q_j - \bar{q}$), momentum p is a $W \times W$ matrix and the leapfrog updates are

$$q \leftarrow q + \epsilon\, p\, C, \qquad p \leftarrow p - \epsilon\, \nabla U(q)\, C^\top,$$

so positions move only within the affine subspace spanned by the ensemble and the effective metric is the empirical ensemble covariance $\Sigma = \mathrm{cov}(C)$. This makes the move *affine-invariant*: it is unchanged under any invertible linear reparametrization of q, so its efficiency is independent of the target's conditioning. (A complementary "h-side" move proposes along a single random ensemble difference direction; both are Metropolis-corrected with the standard HMC accept ratio.) The preconditioner is recomputed from the live ensemble *every iteration* — there is no separately-fit mass matrix.

Adaptation (warmup only). Three independently-toggleable adapters run during warmup and are then frozen:

1. *Initial step size* via a heuristic line search (`find_init_step_size`): double/halve ϵ until the first trajectory's acceptance brackets the target.
2. *Step size* ϵ via Nesterov dual averaging (Hoffman & Gelman 2014) toward a target acceptance (we pin 0.8 for the scaling law). The log-step update uses gain $\sqrt{t}/\gamma$ on the running acceptance error $\bar{H}$; the realized acceptance is the harmonic mean of per-walker $\min(1, e^{\log \alpha})$.

3. *Integration length* $T = \epsilon L$ via the **ChEES** criterion (Hoffman et al. 2021). ChEES does ADAM ascent on the expected change in squared jump distance: the per-step gradient signal is $g \propto h_t\, T\, (\|c_{\text{pro}}\|_\Sigma^2 - \|c_{\text{cur}}\|_\Sigma^2)\, (c_{\text{pro}}^\top \Sigma^{-1} v)$, where c are centered positions, v the trajectory velocity, h_t a Halton sequence, and $\|\cdot\|_\Sigma$ the *affine-invariant* Σ^{-1} norm (matching the move's metric). ADAM (β_1=0, β_2=0.95, lr 0.025, per-step clip $\pm$0.35) updates $\log T$; the realized length is Halton-jittered, $L = \lceil T/\epsilon \rceil$ capped at $L_{\max}$.

Setting all three flags off reduces warmup to a plain burn-in at a fixed (ϵ, L). PEACHES thus realizes *optimally-tuned* ensemble HMC: the metric is free (the ensemble), the step size hits a target acceptance, and the trajectory length maximizes squared jump distance.

Baselines. `emcee` is the gradient-free affine-invariant stretch move (Goodman & Weare 2010; `emcee`, Foreman-Mackey et al. 2013): it is exactly affine-invariant like the h-walk but uses *no* gradient. NUTS is `numpyro`'s implementation with window mass-matrix adaptation; we use the mass matrix appropriate to each experiment (§below) and also an *ablation* with the mass fixed to the identity (`adapt_mass_matrix=False`) to isolate the role of preconditioning.

Targets, metric, convergence. Targets are the isotropic Gaussian $N(0, I_d)$, the rotated ill-conditioned Gaussian (condition number $\kappa = e^{\log \kappa}$), and the Rosenbrock banana — all with exact moments. Efficiency is *production-only* ESS per target-gradient evaluation (ArviZ rank-normalized split ESS, worst coordinate; `emcee` counts one log-density per walker per iteration). Ensemble methods use $n_{\text{walkers}} = \max(128, 4d)$ (so the empirical covariance is full rank) and a stretch pre-warmup (§5). **Every point is checked for convergence**: we plot it solid only if the rank-normalized split $\hat{R} \leq 1.03$ (relaxed slightly from 1.01 for 3-seed-median plus the chain thinning used to make ESS tractable at large N; all genuine failures sit at $\hat{R} \geq 1.04$), and hollow otherwise — a hollow point's ESS/grad is not a meaningful efficiency.

2 Experiment 1 — dimensional scaling

Setup. Target $N(0, I_d)$ (`--no-rotate --log-cond 0`) for $d \in \{8, 16, 32, 64, 128, 256\}$, $n_{\text{walkers}} = 4d$, 3 seeds. PEACHES pins `--target-accept 0.8` so the step-size law holds at a fixed acceptance; warmup 2000, production 3000 (tag `exp4c`), $L_{\max} = 512$. `emcee` mixes far more slowly, so it is run much longer — warmup 2×10^4, production 1×10^5 — and shown only where it reaches $\hat{R} \leq 1.03$ ($d \leq 64$).

Result. Optimally-tuned HMC needs $\epsilon \sim d^{-1/4}$ at fixed acceptance, hence $L \sim d^{1/4}$ and ESS/grad $\sim d^{-1/4}$ (Beskos et al. 2013). PEACHES realizes this: over $d \in [64, 256]$ the fitted slope is -0.235 (theory -0.25) and $\bar{L}$ grows $5.7 \to 9.8$. `emcee`'s gradient-free stretch move scales as ESS/grad $\sim d^{-1}$ (fitted -1.2 over the converged $d \in [8, 64]$); already at d=64 PEACHES is $\sim$230$\times$ more efficient per gradient, and the gap widens exponentially in d.

3 Experiment 2 — conditioning vs. curvature

Setup. (a) Rotated Gaussian d=50, $\kappa = e^{2,4,6,8}$, warmup 1000; PEACHES (production 5000, tag `exp2_lc`k), `emcee` (production 6×10^4), NUTS dense-mass (`exp2_lc`k, the standard sampler) and NUTS identity-mass (`--nuts-mass identity`, production 2000, tag `idmass_lc`k, the ablation). (b) Rosenbrock $d \in \{8, 16, 32\}$ (tag `exp1`): PEACHES (production 8000, $L_{\max} = 256$ for the long

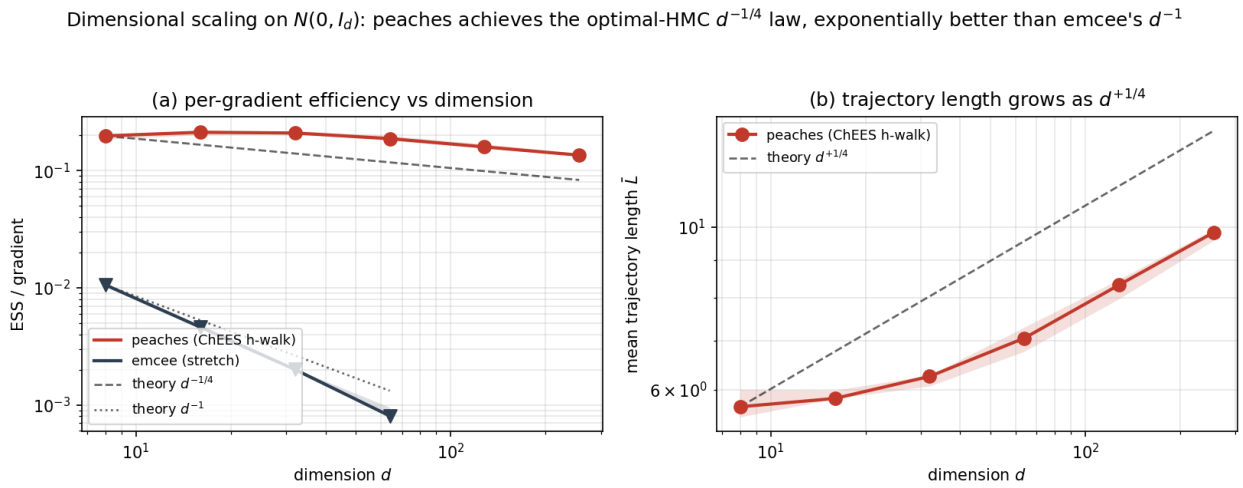


Figure 1: Isotropic Gaussian $N(0, I_d)$. Median over 3 seeds, band = seed min–max. **(a)** per-gradient efficiency: PEACHES tracks the optimal-HMC $d^{-1/4}$ law, `emcee` falls as d^{-1} — an exponentially widening gap. **(b)** the mechanism: PEACHES's ChEES-tuned mean trajectory length grows as $d^{+1/4}$. (NUTS also achieves $d^{-1/4}$ scaling and is omitted here so the peaches-vs-emcee contrast reads cleanly; it is benchmarked in §3–4.)

curved trajectories), `emcee` (production 10^5), NUTS identity-mass (`idmass`). On Rosenbrock the global conditioning is mild, so mass adaptation is irrelevant and the identity-mass NUTS represents NUTS faithfully.

Result — a clean 2×2. The two targets isolate two distinct failure modes. *Conditioning* (panel a): the empirical-covariance metric makes PEACHES and `emcee` affine-invariant, so their ESS/grad is flat in κ ($\approx$0.19 and $\approx 10^{-3}$, the latter just dimensionally penalised at d=50). NUTS is flat in κ *only* through its mass adaptation: with the mass fixed to the identity its ESS/grad falls from 3.1×10^{-2} ($\kappa = e^2$) to 3.0×10^{-6} ($\kappa = e^8$) — four orders, and unconverged ($\hat{R} = 1.7$) at $\kappa = e^8$. *Curvature* (panel b): Rosenbrock's position-dependent curvature defeats any fixed linear map. `emcee`'s single affine stretch move cannot cross the ridge — even at 10^5 samples it fails to converge for $d \geq 4$ ($\hat{R}$ grows $1.14 \to 2.0$ over d=8$\to$32). PEACHES and NUTS integrate *local* Hamiltonian dynamics along the ridge and stay converged (ESS/grad $\approx 10^{-3}$). Only PEACHES clears both axes: the ensemble preconditioner gives it the conditioning-robustness un-adapted NUTS lacks, and gradient dynamics give it the curvature-handling `emcee` lacks.

4 Experiment 3 — warmup robustness

Setup. Ill-conditioned Gaussian (d=100, $\kappa = e^6$) and Rosenbrock (d=8); warmup $\in \{50, 100, 200, 500, 1000\}$, production 5000, 3 seeds (tag `exp3`). NUTS uses a dense mass matrix (the standard choice on an ill-conditioned target). PEACHES uses a 200-step stretch pre-warmup (§5) then its own warmup of the given length.

Result. Because the preconditioner is recomputed from the live ensemble every step, PEACHES needs no mass-matrix warmup: its ESS/grad is flat at $\approx$0.17 (Gaussian) across all warmup lengths. NUTS-dense must fit a $d(d{+}1)/2$-entry dense mass during warmup; at warmup 50 its ESS/grad is

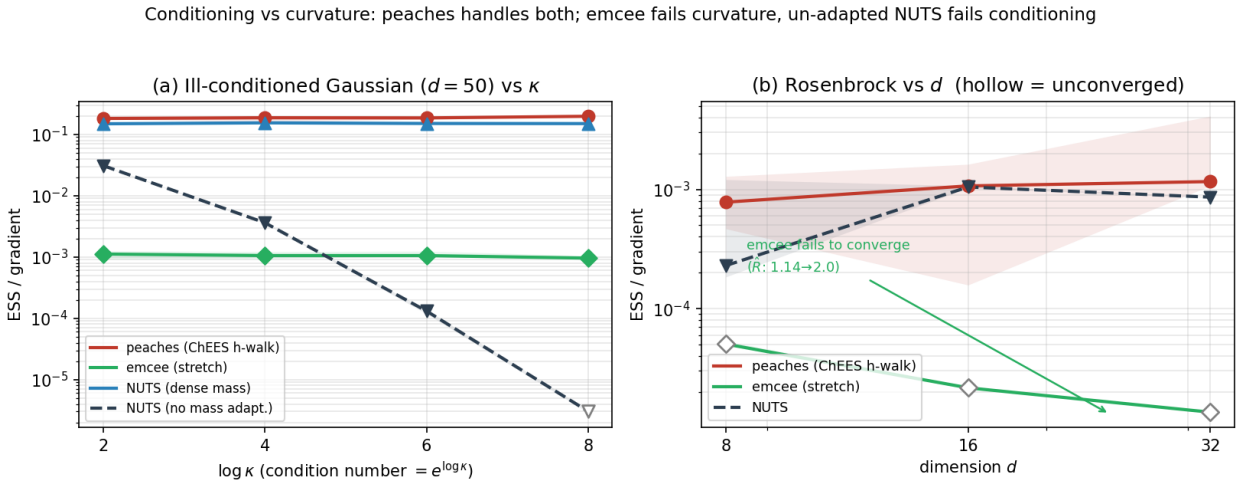


Figure 2: Median over 3 seeds, band = seed min–max; solid = converged ($\hat{R} \leq 1.03$), hollow = not. **(a)** ill-conditioned Gaussian (d=50) vs. κ: PEACHES/`emcee`, and NUTS *with* dense-mass adaptation, are flat in κ; NUTS with the mass fixed to the identity collapses 4 orders. **(b)** Rosenbrock vs. d: PEACHES and NUTS (gradient methods) cross the curved ridge; `emcee` fails to converge for $d \geq 4$.

3.7×10^{-5} — a factor ~ 5000 below PEACHES — and it becomes competitive only once warmup ≥ 500. The same pattern holds on Rosenbrock. This is the practical face of "the metric is free": the work NUTS spends in a long, target-dependent warmup, PEACHES gets for nothing.

5 Practical recommendation: a three-phase run

For PEACHES (and ensemble HMC generally) we recommend the following workflow, which the benchmarks above all use:

1. **Stretch pre-warmup (gradient-free).** Run a short `emcee`-style stretch move for w_s iterations *first*, from an over-dispersed initialization. This is cheap (one log-density per walker, no gradient) and affine-invariant, and it does two essential things: it moves the ensemble *mean* onto the typical set (a reasonable centre), and it shapes the ensemble *covariance* to the target's local geometry. This matters because PEACHES's preconditioner *is* the empirical covariance: an isotropic or collapsed initial cloud gives an ill-shaped metric, which in turn collapses the dual-averaged step size and wastes the gradient warmup. A few hundred stretch steps ($w_s \approx 200$; benefit saturates quickly) suffice; it uses no gradient and no knowledge of the target covariance.

2. **peaches warmup (gradient).** With the ensemble now well-centred and well-shaped, run the PEACHES adapters: `find_init_step_size` for the initial ϵ, dual averaging to the target acceptance, and ChEES for the integration length. The empirical-covariance metric is re-estimated each step, so no mass matrix is fit here — only the scalar ϵ and the length L.

3. **Production.** Freeze (ϵ, L) and the adapters and collect samples. ESS/grad is measured here only.

In the code this is exactly (`warm_start` w_s`,` `warmup` w`,` `num_samples` N). Phase 1 is the ensemble analogue of NUTS's mass-matrix warmup, but gradient-free and target-agnostic; phase

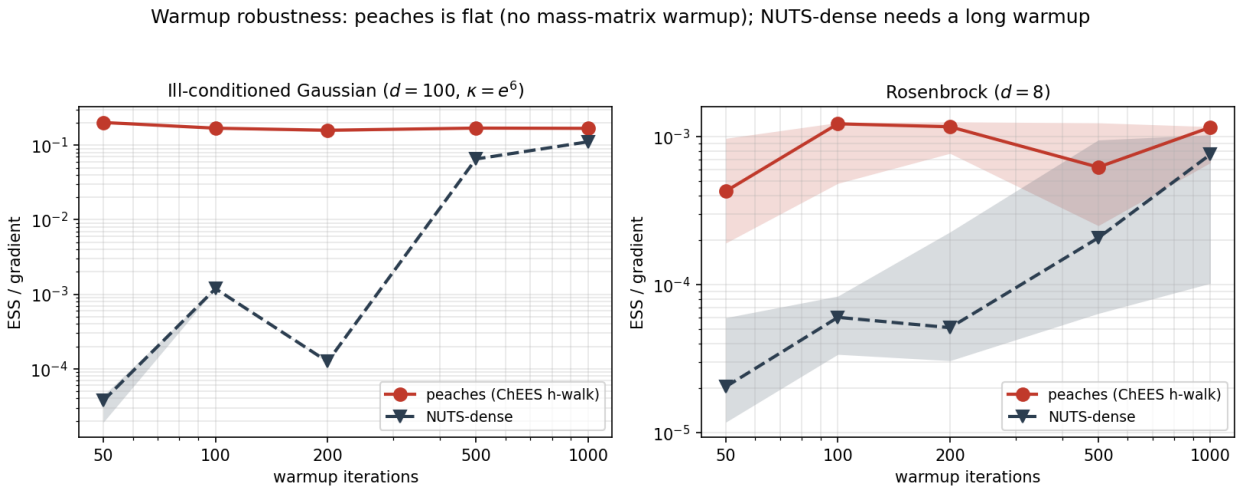


Figure 3: ESS/grad vs. gradient-sampler warmup length, 3 sampling seeds on a fixed target (median, band = seed min–max), production 5000. PEACHES is flat from warmup 50 to 1000; NUTS-dense needs $\gtrsim$500 warmup steps to fit its mass matrix and is orders less efficient below that.

2 is much shorter and more robust than NUTS's because there is no mass matrix to converge (§4). Skipping phase 1 is the most common failure mode: without it the worst-mixing direction is initialization-dependent and the step size can collapse.

6 Reproducibility

All runs go through `python -m benchmark.run` (one `(target,dim,sampler,seed)` cell); the experiment driver `benchmark.experiments` enumerates the grids. Key flags: `--target` {`gaussian,rosenbrock`}, `--dim`, `--sampler` {`peaches,stretch,nuts`}, `--log-cond` ($\log\kappa$; 0 + `--no-rotate` for isotropic), `--n-chains`, `--warm-start` (stretch pre-warmup), `--warmup`, `--num-samples`, `--target-accept`, `--max-L` (PEACHES), `--nuts-mass` {`dense,diag,identity`}, `--max-tree-depth`, `--seed`, `--tag`.

Experiment	target	sampler configs	warmup/prod.
Scaling (§2)	$N(0, I_d)$, $d \leq 256$	PEACHES $L_{\max}$=512; `emcee`	2k/3k; emcee 20k/
Conditioning (§3a)	Gauss d=50, κ=$e^{2\text{-}8}$	PEACHES; `emcee`; NUTS `dense/identity`	1k/5k; emcee /60k
Curvature (§3b)	Rosenbrock $d \in \{8, 16, 32\}$	PEACHES $L_{\max}$=256; `emcee`; NUTS `identity`	5k/8k; emcee /10^5
Warmup (§4)	Gauss d=100 κ=e^6; Rosen d=8	PEACHES (w_s=200); NUTS `dense`	{50..1000}/5k

All targets use 3 seeds; $n_{\text{walkers}} = \max(128, 4d)$. Convergence ($\hat{R}$), bias, and std-error are stored alongside ESS/grad in each `metrics.json`. Figures are produced by `benchmark/plot_hwm_*.py`.

References. Beskos, Pillai, Roberts, Sanz-Serna, Stuart (2013), *Optimal tuning of the hybrid Monte Carlo algorithm*, Bernoulli 19(5A). Goodman & Weare (2010), *Ensemble samplers with affine invariance*, Comm. App. Math. Comp. Sci. 5(1). Foreman-Mackey, Hogg, Lang, Goodman (2013), *emcee: the MCMC hammer*, PASP 125. Hoffman & Gelman (2014), *The No-U-Turn Sampler*, JMLR 15. Hoffman, Radul, Sountsov (2021), *An adaptive-MCMC scheme for setting trajectory lengths in Hamiltonian Monte Carlo*, AISTATS, PMLR 130.